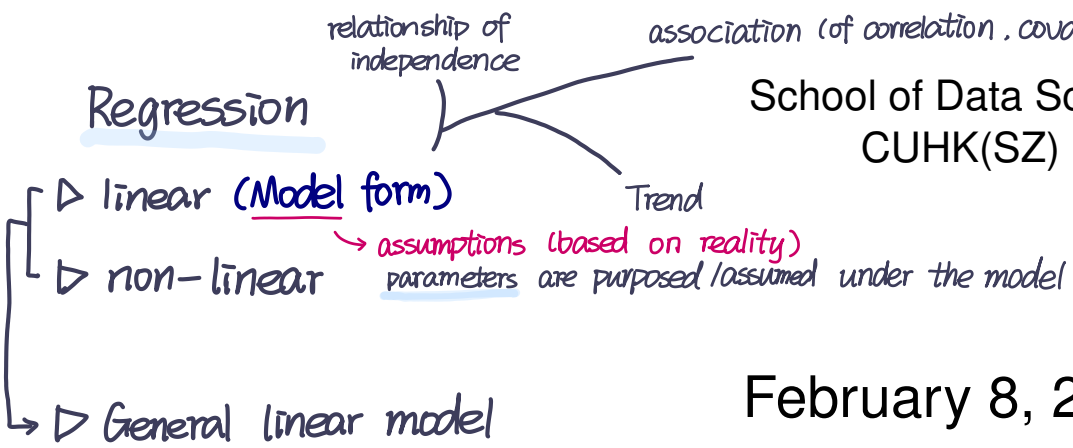
STA2002: Probability and Statistics II
Introduction to Regression

Pay attention to what you want to anaylsis. ☆

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[Another tool: method of moment]
↓
MLE: one of the tools to estimate parameters

▷ we have unknow parameters in model

▷ we believe true value of parameters exist.

▷ want to estimate the true value.

Introduction

- In this lecture we will begin our discussion on simple linear regression model.
  As often there is interest in the relation between two variables. For example, the temperature at which a certain chemical reaction is performed and the yield of a chemical compound resulting from the reaction. Frequently, one of these variables, say, x, is known in advance of the other, so there is interest in predicting a future random variable Y.
- **Suggested reading**: Chapter 6.5, 7.6 of the textbook.

usage of statistics

eg. Esitimation of parameters in a statistical model

for instance, linear regression model.

Introduction

- We have two variables, namely X and Y, and we would like to study the relationship between these two variables.
- X is often called explanatory/predictor/independent variable, and Y is often called the response/dependent variable.
- Example: Y is exam score, and X is the amount of time spent on study.

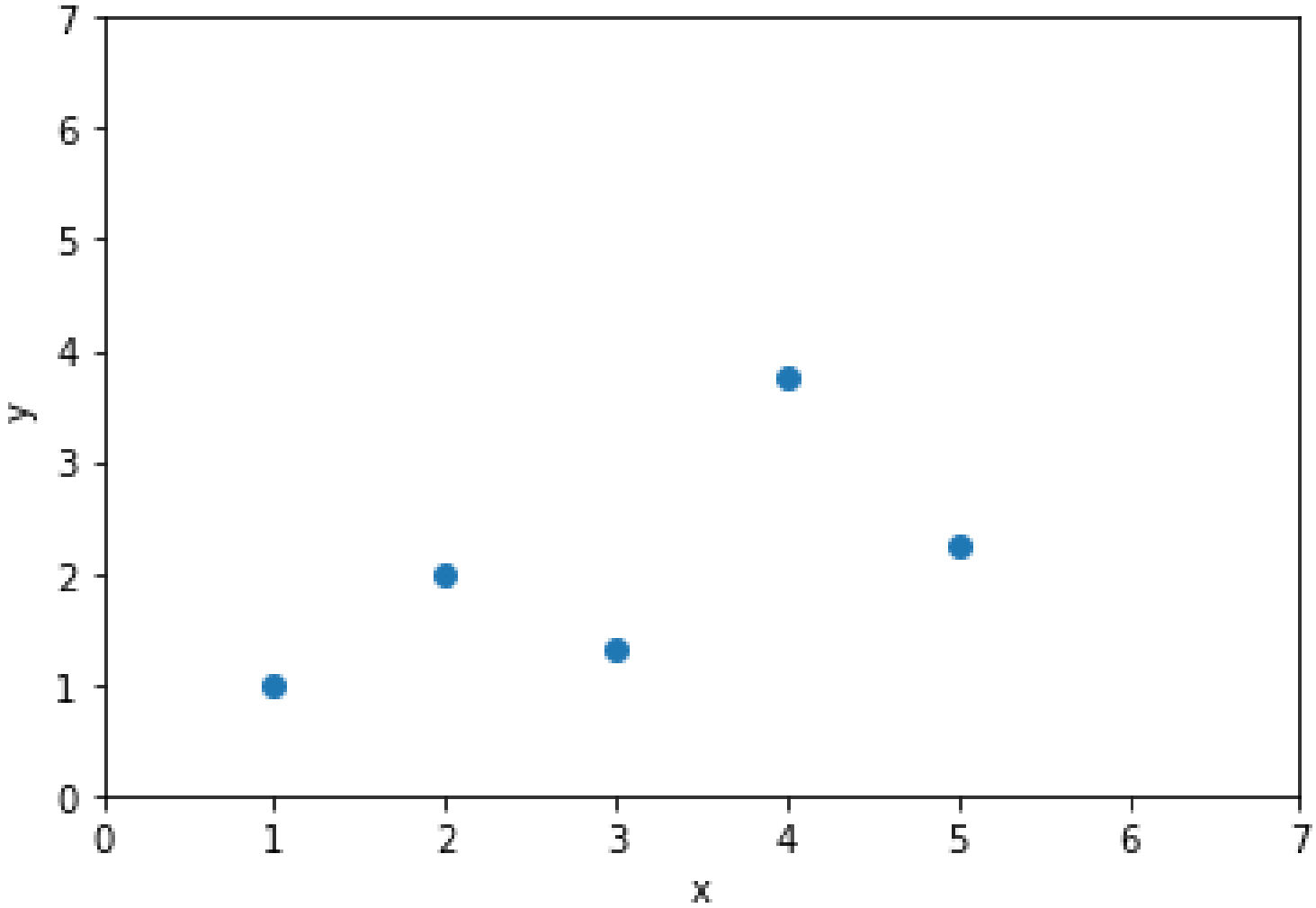
Data Example

- Here we use the following data as our running example,

x_i	y_i
1.0	1.0
2.0	2.0
3.0	1.3
4.0	3.75
5.0	2.25

- Plot these data in a scatter plot.

Data Example: Scatter Plot



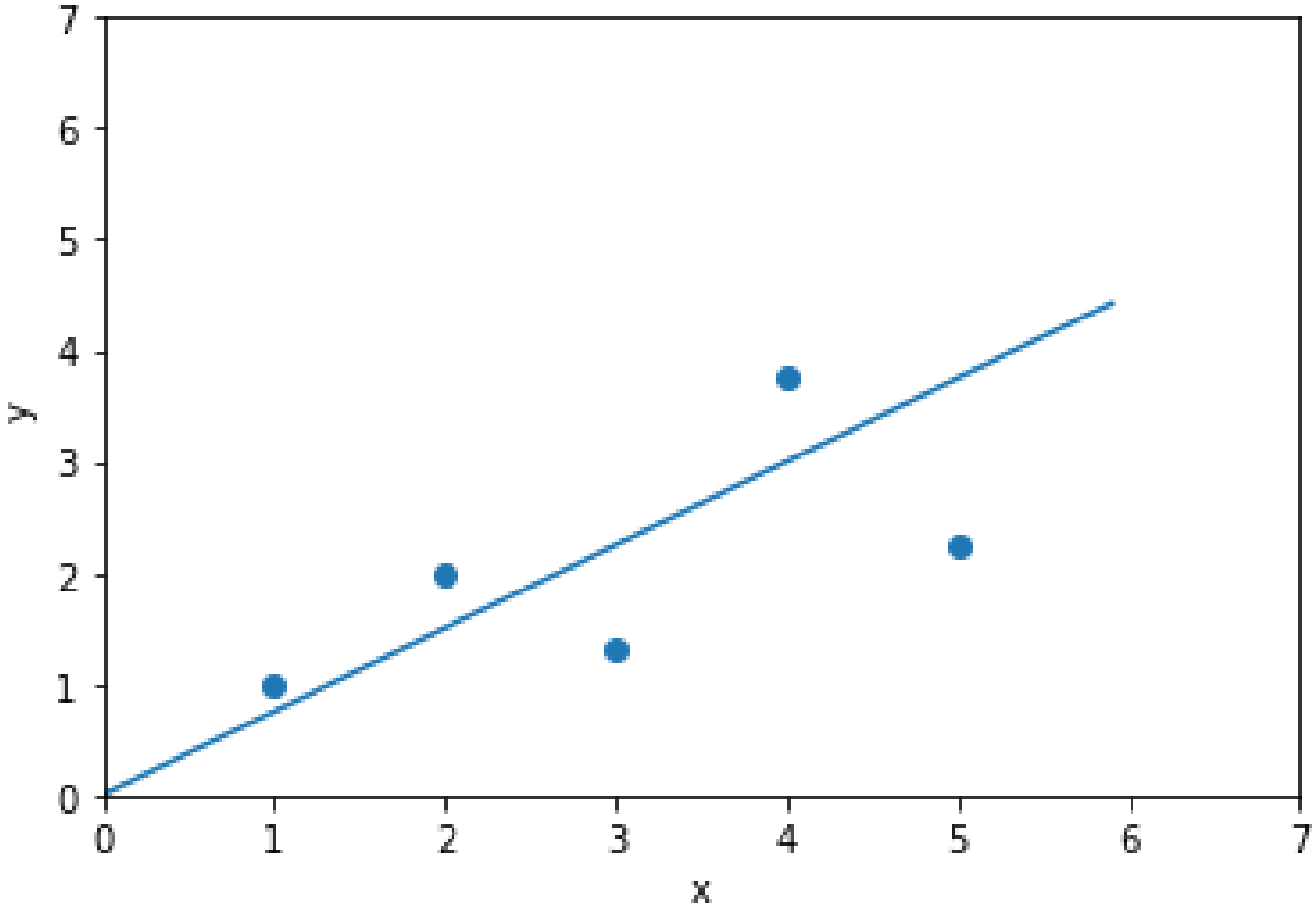
Data Example

- The plot looks linear, suggesting perhaps we can use a linear equation of the form $\hat{y} = \alpha + \beta x$ to predict y using x.
- For example, if we use the line $\hat{y} = 0 + 0.75x$. The new table is

x_i	y_i	$\hat{y}_i$
1.0	1.0	0.75
2.0	2.0	1.50
3.0	1.3	2.25
4.0	3.75	3.00
5.0	2.25	3.75

- We also overlay our scatter plot with this $\hat{y}$.

Data Example: Fitted Scatter Plot



Least Squares Regression Line

- The difference $y_i - \hat{y}_i$ is called the ith **residuals.**
- While there are many lines of the form $\hat{y} = \alpha + \beta x$ that we can choose, we would like the sum of the squared residuals of $\hat{y}$ to be as small as possible.
- We want to find the best line that **minimize the sum of the squared residuals**.
- In other words, we would like to solve

$$\min_{\alpha,\,\beta} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \min_{\alpha,\,\beta} \sum_{i=1}^{n} (y_i - (\alpha + \beta x_i))^2.$$

Least Squares Regression Line

Definition 1 (Least Squares Regression Line)

The least squares regression line is the line of the form $\hat{y} = \alpha + \beta x$, where α, β solve

$$\min_{\alpha,\,\beta} \sum_{i=1}^{n} (y_i - \hat{y}_i)^2 = \min_{\alpha,\,\beta} \sum_{i=1}^{n} (y_i - (\alpha + \beta x_i))^2.$$

- Question: How to solve the optimization problem proposed above?

Transformation of the Problem

- Follow the book notation, instead of considering lines of the form

$$\hat{y}_i = \alpha^{old} + \beta x_i,$$

now we consider lines of the form

$$\hat{y}_i = \alpha^{new} + \beta(x_i - \overline{x}),$$

where $\alpha^{new} - \beta\overline{x} = \alpha^{old}$, and $\overline{x}$ is the sample mean.

- Now, we would like to solve the following equivalent optimization problem,

$$H(\alpha,\beta) \quad \min_{\alpha,\,\beta} \sum_{i=1}^{n} (y_i - \alpha - \beta(x_i - \overline{x}))^2.$$

$$\frac{\partial H(\alpha,\beta)}{\partial \alpha} = -2\sum (y_i - \alpha - \beta(x_i - \bar{x})) = 0$$

$$\Rightarrow \alpha = \frac{\sum [y_i - \beta(x_i - \bar{x})]}{n} = \frac{\sum y_i}{n} - \beta \underbrace{\frac{\sum (x_i - \bar{x})}{n}}_{0} = \bar{y}$$

$$\frac{\partial H(\alpha,\beta)}{\partial \beta} =$$

Problem Solution

- Let the objective function be denoted as H,

$$H(\alpha, \beta) := \sum_{i=1}^{n} (y_i - \alpha - \beta(x_i - \overline{x}))^2.$$

- Take the partial derivatives of H with respect to α and β,

$$\frac{\partial H(\alpha,\beta)}{\partial \alpha} = 2 \sum_{i=1}^{n} (y_i - \alpha - \beta (x_i - \overline{x}))(-1) \overset{\text{set}}{=} 0.$$

$$\frac{\partial H(\alpha,\beta)}{\partial \beta} = 2 \sum_{i=1}^{n} (y_i - \alpha - \beta (x_i - \overline{x}))(-(x_i - \overline{x})) \overset{\text{set}}{=} 0.$$

Problem Solution

- Simplifying the expressions yields,

$$\begin{aligned}\sum_{i=1}^{n} y_i &= n\alpha + \beta \sum_{i=1}^{n} (x_i - \overline{x}) \\ &= n\alpha + 0 \\ &= n\alpha,\end{aligned}$$

and so,

$$\hat{\alpha} = \frac{1}{n} \sum_{i=1}^{n} y_i = \overline{y}.$$

Problem Solution

- And also,

$$\sum_{i=1}^{n} y_i (x_i - \overline{x}) = \sum_{i=1}^{n} (\alpha + \beta (x_i - \overline{x})) (x_i - \overline{x})$$
$$= \alpha \sum_{i=1}^{n} (x_i - \overline{x}) + \beta \sum_{i=1}^{n} (x_i - \overline{x})^2$$
$$= \beta \sum_{i=1}^{n} (x_i - \overline{x})^2,$$

and so

$$\hat{\beta} = \frac{\sum_{i=1}^{n} y_i (x_i - \overline{x})}{\sum_{i=1}^{n} (x_i - \overline{x})^2}.$$

Problem Solution

- **The least squares regression line** is

$$\hat{y}_i = \alpha_{LS} + \beta_{LS}(x_i - \overline{x}),$$

where

$$\alpha_{LS} = \overline{y},$$

$$\beta_{LS} = \frac{\sum_{i=1}^{n} y_i (x_i - \overline{x})}{\sum_{i=1}^{n} (x_i - \overline{x})^2}.$$

A Simple Linear Regression Model

- We now would like to propose a statistical model, where α and β can be treated as parameters and we can discuss their confidence intervals or to conduct further statistical tests.

Definition 2 (Simple linear regression model)

Let $\epsilon_1, \ldots, \epsilon_n \overset{i.i.d.}{\sim} N(0, \sigma^2)$. Let

3 parameters: α, β, δ

$$Y_i = \alpha + \beta(x_i - \overline{x}) + \epsilon_i.$$

Y_i: R.V [Randomness of Y_i is putted in ϵ_i]

y_i is realization.

Note that x_i for $i = 1, \ldots, n$ and $\overline{x}$ are fixed and deterministic values, while Y_i for $i = 1, \ldots, n$ are mutually independent random variables with mean $\alpha + \beta(x_i - \overline{x})$ and variance σ^2.

Theorem 3 (MLEs of α, β and σ^2)

Suppose that $Y_1, \ldots, Y_n$ are independent random samples observed from the simple linear regression model. The maximum likelihood estimators of α, β and σ^2 are

MLE

$$\hat{\alpha} = \alpha_{LS} = \overline{Y},$$

$$\hat{\beta} = \beta_{LS} = \frac{\sum_{i=1}^{n} Y_i (x_i - \overline{x})}{\sum_{i=1}^{n} (x_i - \overline{x})^2},$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^{n} \left[Y_i - \hat{\alpha} - \hat{\beta} (x_i - \overline{x}) \right]^2.$$

Note that $\hat{\alpha}, \hat{\beta}$ and $\hat{\sigma}^2$ are all random variables in $Y_1, \ldots, Y_n$.

key idea: maximize likelihood function with parameters unknown

MLE : maximum likelihood estimation [point estimation]

↔ Confidence interval [interval estimation]

▷ likelihood function of the model

e.g. $Y_i = \alpha + \beta(X_i - \bar{X}) + \varepsilon_i \quad i = 1, 2 \dots n$

assume $\varepsilon_i \sim N(0, \delta^2) \quad i = 1, 2 \dots n$, MLE?

Hint: under this model, likelihood func. is just the joint distribution of Y_i $(i = 1, 2 \dots n)$

MLE is a statistics.

e.g.: $\hat{\alpha}$ $\hat{\beta}$ $\hat{\delta}$

Statistics are R.Vs.

⇒ Distribution of MLE can be studied.

Proof of the Theorem

- Proof. The likelihood function is

the joint distribution of R.Vs (Y_i here) independent.

mean $\bar{Y}$

$$L\left(\alpha,\beta,\sigma^2\right)=\prod_{i=1}^{n}\frac{1}{\sqrt{2\pi\sigma^2}}\exp\left\{-\frac{\left[y_i-\alpha-\beta\left(x_i-\bar{x}\right)\right]^2}{2\sigma^2}\right\}$$

Var(Y)

$$=\left(\frac{1}{2\pi\sigma^2}\right)^{n/2}\exp\left\{-\frac{\sum_{i=1}^{n}\left[y_i-\alpha-\beta\left(x_i-\bar{x}\right)\right]^2}{2\sigma^2}\right\}.$$

- The log-likelihood function is

$=$ monotonic (simplify the function)

$$l(\alpha,\beta,\sigma^2)=-(n/2)\log(2\pi\sigma^2)-\frac{H(\alpha,\beta)}{2\sigma^2}.$$

Proof of the Theorem

- Take the partial derivatives of l with respect to α, β, σ^2,

$$\frac{\partial l(\alpha, \beta, \sigma^2)}{\partial \alpha} \overset{\text{set}}{=} 0 \quad \Rightarrow \quad \frac{\partial H(\alpha, \beta)}{\partial \alpha} = 0$$
$$\Rightarrow \quad \hat{\alpha} = \alpha_{LS} = \overline{Y},$$

$$\frac{\partial l(\alpha, \beta, \sigma^2)}{\partial \beta} \overset{\text{set}}{=} 0 \quad \Rightarrow \quad \frac{\partial H(\alpha, \beta)}{\partial \beta} = 0$$
$$\Rightarrow \quad \hat{\beta} = \beta_{LS} = \frac{\sum_{i=1}^{n} Y_i (x_i - \overline{x})}{\sum_{i=1}^{n} (x_i - \overline{x})^2},$$

Proof of the Theorem

- And also,

$$\frac{\partial l(\alpha,\beta,\sigma^2)}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{\sum_{i=1}^{n}\left[y_i - \alpha - \beta\left(x_i - \overline{x}\right)\right]^2}{2\left(\sigma^2\right)^2} \overset{\text{set}}{=} 0,$$

treat δ^2 as parameter (not δ)

$$\Rightarrow \quad \hat{\sigma}^2 = \frac{1}{n}\sum_{i=1}^{n}\left[Y_i - \hat{\alpha} - \hat{\beta}\left(x_i - \overline{x}\right)\right]^2.$$

$(\hat{\delta})^2 = \hat{\delta}^2$

- **Remark**. Theoretically, these solutions are the unique global maximizers of $l(\alpha,\beta,\sigma^2)$. □

Further Remark

- Some useful computational formulas for the maximum likelihood *estimates* are

$$\hat{\alpha} = \frac{1}{n}\sum_{i=1}^{n} y_i = \overline{y},$$

$$\hat{\beta} = \frac{\sum_{i=1}^{n} x_i y_i - n\overline{xy}}{\sum_{i=1}^{n} x_i^2 - n\overline{x}^2},$$

$$\widehat{\sigma^2} = \frac{1}{n}\left(\sum_{i=1}^{n} y_i^2\right) - \overline{y}^2 - \hat{\beta}\cdot\frac{1}{n}\left(\sum_{i=1}^{n} x_i y_i\right) + \hat{\beta}\overline{xy}.$$

Data Example Again

- Returning to our running example, we have

$$\sum_{i=1}^{n} x_i = 15, \quad \sum_{i=1}^{n} y_i = 10.3,$$

$$\sum_{i=1}^{n} x_i^2 = 55, \quad \sum_{i=1}^{n} y_i^2 = 25.815, \quad \sum_{i=1}^{n} x_i y_i = 35.15.$$

- As a result, $\hat{\alpha} = 2.06, \hat{\beta} = 0.425, \hat{\sigma}^2 = 0.55815$. The regression line is,

$$y = \hat{\alpha} + \hat{\beta}(x - \overline{x}) = 2.06 + 0.425(x - 3) = 0.785 + 0.425x.$$

Distribution of $\hat{\alpha}, \hat{\beta}$ and $\hat{\sigma}^2$

- To construct confidence intervals for α and β, we need to know the distributions of the mles $\hat{\alpha}, \hat{\beta}$ and $\hat{\sigma}^2$.
- Recall the following theorem in Chapter 5.

Theorem 4 (Theorem 5.5-1 in book)

Let $Y_1, \ldots, Y_n$ be n mutually independent normal random variables with means $\mu_1, \ldots, \mu_n$ and variances $\sigma_1^2, \ldots, \sigma_n^2$ respectively. Then the linear function

$$\sum_{i=1}^{n} c_i Y_i \sim N\left(\sum_{i=1}^{n} c_i \mu_i, \sum_{i=1}^{n} c_i^2 \sigma_i^2\right).$$

Lemma 5

$$\hat{\alpha} = \overline{Y} \sim N\left(\alpha, \frac{\sigma^2}{n}\right)$$

Proof. For each i, $E(Y_i) = \alpha + \beta(x_i - \overline{x})$ and $\mathrm{Var}(Y_i) = \sigma^2$. We take $c_i = \frac{1}{n}$ in Theorem 4, and we have

$$E(\hat{\alpha}) = \frac{1}{n}\sum_{i=1}^{n} E(Y_i) = \alpha + \frac{\beta}{n}\sum_{i=1}^{n}(x_i - \overline{x}) = \alpha.$$

$$\mathrm{Var}(\hat{\alpha}) = \frac{\sigma^2}{n}.$$

□

Lemma 6

$$\hat{\beta} = \frac{\sum_{i=1}^{n} Y_i (x_i - \overline{x})}{\sum_{i=1}^{n} (x_i - \overline{x})^2} \sim N\left(\beta, \frac{\sigma^2}{\sum_{i=1}^{n} (x_i - \overline{x})^2}\right)$$

Proof. We take

$$c_i = \frac{(x_i - \overline{x})}{\sum_{i=1}^{n} (x_i - \overline{x})^2}$$

in Theorem 4, and we have

$$E(\hat{\beta}) = \sum_{i=1}^{n} \frac{(x_i - \overline{x})(\alpha + \beta(x_i - \overline{x}))}{\sum_{i=1}^{n} (x_i - \overline{x})^2} = 0 + \beta \frac{\sum_{i=1}^{n} (x_i - \overline{x})^2}{\sum_{i=1}^{n} (x_i - \overline{x})^2} = \beta.$$

$$\mathrm{Var}(\hat{\beta}) = \sum_{i=1}^{n} \frac{(x_i - \overline{x})^2 \sigma^2}{\left(\sum_{i=1}^{n} (x_i - \overline{x})^2\right)^2} = \frac{\sigma^2}{\sum_{i=1}^{n} (x_i - \overline{x})^2}.$$

□

Lemma 7

$$\frac{n\hat{\sigma}^2}{\sigma^2} = \frac{1}{\sigma^2}\sum_{i=1}^{n}\left(Y_i - \hat{\alpha} - \hat{\beta}\left(x_i - \overline{x}\right)\right)^2 \sim \chi^2(n-2).$$

(Remark. Proof of this theorem is not required.)

Proof. First, note that we have

$$Y_i \sim N(\alpha + \beta(x_i - \overline{x}), \sigma^2)$$

$$\sum_{i=1}^{n}\left(\frac{Y_i - \alpha - \beta\left(x_i - \overline{x}\right)}{\sigma}\right)^2 \sim \chi^2(n)$$

$$\hat{\alpha} \sim N\left(\alpha, \frac{\sigma^2}{n}\right)$$

And also,

$$\left(\frac{\hat{\alpha}-\alpha}{\sigma/\sqrt{n}}\right)^2 \sim \chi^2(1)$$

$$\hat{\beta} \sim N\left(\beta, \frac{\sigma^2}{\sum_{i=1}^{n}(x_i-\overline{x})^2}\right)$$

$$\frac{\left(\hat{\beta}-\beta\right)^2}{\sigma^2/\sum_{i=1}^{n}\left(x_i-\overline{x}\right)^2} \sim \chi^2(1)$$

- Conduct the following expression manipulation,

$$\sum_{i=1}^{n}\left(\frac{Y_i-\alpha-\beta\left(x_i-\overline{x}\right)}{\sigma}\right)^2$$

$$= \frac{1}{\sigma^2}\sum_{i=1}^{n}\left(Y_i-\hat{\alpha}-\hat{\beta}\left(x_i-\overline{x}\right)\right)^2+\frac{n}{\sigma^2}(\hat{\alpha}-\alpha)^2$$

$$+\frac{\sum_{i=1}^{n}\left(x_i-\overline{x}\right)^2}{\sigma^2}(\hat{\beta}-\beta)^2.$$

- Together with Theorem 8 and the property that $\hat{\alpha}$, $\hat{\beta}$ and $\hat{\sigma}^2$ are independent (see Section 7.6 of the textbook), this allows us to conclude that

$$\frac{n\hat{\sigma}^2}{\sigma^2}=\frac{1}{\sigma^2}\sum_{i=1}^{n}\left(Y_i-\hat{\alpha}-\hat{\beta}\left(x_i-\overline{x}\right)\right)^2\sim\chi^2(n-2).$$

□

Further Remark

- Note that

$$E(\hat{\sigma}^2) = \frac{\sigma^2}{n} E\left(\frac{n\hat{\sigma}^2}{\sigma^2}\right) = \frac{n-2}{n}\sigma^2,$$

  thus, $\hat{\sigma}^2$ is a biased estimator of σ^2.

- Meanwhile,

$$S_R^2 = \frac{n\hat{\sigma}^2}{n-2} = \frac{1}{n-2}\sum_{i=1}^{n}\left(Y_i - \hat{\alpha} - \hat{\beta}\left(x_i - \overline{x}\right)\right)^2$$

  is an unbiased estimator of σ^2.

- Equivalently,

$$\frac{(n-2)S_R^2}{\sigma^2} \sim \chi^2(n-2).$$

An Useful Theorem

Theorem 8 (Theorem 9.3.1)

Let $Q = Q_1 + Q_2 + \cdots + Q_k$, where $Q, Q_1, ..., Q_k$ are $k + 1$ real quadratic forms in n mutually independent random variables normally distributed with the same variance σ^2.

Let $Q/\sigma^2, Q_1/\sigma^2, \ldots, Q_{k-1}/\sigma^2$ have chi-square distributions with $r, r_1, ..., r_{k-1}$ degrees of freedom, respectively. If Q_k is nonnegative, then

(a) $Q_1, \ldots, Q_k$ are mutually independent, and hence,
(b) Q_k/σ^2 has a chi-square distribution with $r - (r_1 + \cdots + r_{k-1}) = r_k$ degrees of freedom.

The proof is not required.